

Deliverable 1

Tutorial can be found [here](#)

Concepts I don't understand:

- non-essential ports: network ports that are not critical for a system's core functionality or are not commonly used for essential services
- Apache: The web server software that processes requests and serves web assets and content via HTTP
- domain: an administrative structure for organizing, delivering and accessing services on the internet.
- web root: the main folder on a web server where a website's files, like HTML, CSS, and images, are stored
- virtual host file: allows a single physical server to host multiple websites or services, each with its own unique domain name or IP address, without requiring separate physical machines.

What is a web server? Hardware and software side

A web server on the hardware side is a physical computer that stores website files such as HTML, CSS, images, and JavaScript. It connects to the Internet to serve content when requested by users. On the software side, a web server uses HTTP server software that understands URLs (web addresses), follows HTTP/HTTPS protocols, and sends website content to users' browsers.

What are some different web server applications?

Apache HTTP Server (open-source, widely used)  <https://www.apache.org/img/asf-estd-1999-logo.jpg>

Nginx (lightweight, high performance)

Microsoft IIS (Internet Information Services, Windows-based)

LiteSpeed (commercial, fast)

Caddy (automatic HTTPS)

What is virtualization?

Virtualization is the process of creating a virtual version of physical components like operating systems, servers, storage devices, or networks. It allows multiple operating systems or applications to run on a single physical machine, improving efficiency and flexibility.

What is virtualbox?

VirtualBox is a free, open-source virtualization software developed by Oracle. It allows users to create and run virtual machines (VMs) on their current system, supporting different operating systems such as Windows, Linux, and macOS simultaneously.

What is a virtual machine?

A virtual machine is a software-based emulation of a physical computer. It runs its own operating system and applications, operating completely independently and isolated from the host system.

What is Ubuntu Server?

Ubuntu Server is the server edition of the Ubuntu operating system. It is typically command-line only, meaning it does not come with a graphical user interface (GUI) by default. Ubuntu Server is focused on providing high stability, security, and the ability to host services like websites, databases, and cloud platforms.

What is a firewall?

A firewall is a security system, either hardware- or software-based, that monitors and filters network traffic. It blocks or allows data packets based on specific security rules to protect computers and networks from unauthorized access and cyberattacks.

What is SSH?

SSH (Secure Shell) is a network protocol that provides encrypted access to remote computers. It is commonly used for remote server management and secure file transfers (such as with SCP or SFTP), and it typically operates over port 22.